LR-GoWU: Learned-Return Go-With-Uncertainty

Pseudo-algorithms (iteration v2, commit `d33407c`)

Idea. GoWU [1] redistributes particles by *restoring* emulator states ($\textsc{Reset}(p_{\text{target}}, p_{\text{source}})$). LR-GoWU keeps GoWU's winner selection, group consolidation, global synchronization and rollback, but never restores a state: every episode starts at $s_0 = \mathcal{E}.\textsc{reset}()$ and a learned goal-conditioned *return policy* π_R walks to the selected landmark. Exploration with GoWU's particle policy starts only after this *handoff*. The search tree of checkpoints becomes an archive of *landmarks*, one per discretized cell. The archive supplies hindsight demonstrations that π_R imitates. The output is a deployable policy (run π_R from reset), not only a trajectory. The primed numbers mark the counterpart of the paper's algorithm.

GoWU (paper)	**LR-GoWU (here)**
particle p_i on a simulator checkpoint	episode slot i, always started from s_0
$\textsc{Reset}(p_i, v)$	return phase: π_R walks to landmark g (Alg. 1′, l. 6–21)
lineage tree $\mathcal{T}$, node $v = (s_v, R_v, u_v)$	landmark archive $\mathcal{A}$, one landmark per cell (Alg. 2′)
outer step of T_i inner steps	explore phase after handoff (Alg. 1′, l. 21–31)
$\mathcal{T}.\textsc{AddChild}(p_i)$	$\textsc{UpdateArchive}$ on every alive step
winner reset of $\textsc{Dead}$ particles	next goal of a dead slot ← group winner (Alg. 4′, l. 17)
rollback when *all* particles die	rollback when returns to a landmark keep failing (Alg. 3′)

Terms and definitions

cell(s)	Discretization of a state, read from RAM: (level, room, #keys held, $\lfloor x/16 \rfloor$, $\lfloor y/16 \rfloor$).
Landmark c	The best state seen so far in one cell. It stores the frame f_c (84×84 grey), the cumulative extrinsic reward R_c of the episode when it was stored, the uncertainty U_c, the parent landmark u_c, the depth d_c and the return statistics below. Landmarks are replaced, never removed.
Archive $\mathcal{A}$	The set of landmarks. Its root $c_0 = \text{cell}(s_0)$ has $R_{c_0} = 0$.
$U(f)$	Epistemic uncertainty: RND prediction error on one frame, shared by all slots. Stored values U_c are *re-scored* with the current RND whenever they enter a comparison.
$\succ^{\text{lex}}$, lex-argmax	As in the paper: $(b, c) \succ^{\text{lex}} (b', c')$ iff $b > b'$ or ($b = b'$ and $c > c'$). lex-argmax$_{c \in S}(R_c, U_c)$ is the landmark with the highest R, and among those the highest re-scored U.
DEAD	A step is DEAD if a life is lost, the game is over, or the exploit guard fires (the middle-room key bug; its reward is zeroed). A DEAD step ends the episode. All other steps are ALIVE.
REACH(s, R, g)	cell(s) $= g$ and $R \geq R_g$: the agent stands in g's cell with at least g's reward.
Anc$_k(g)$	The k-th ancestor of g via parent pointers (Anc$_1(g) = u_g$), clamped to the farthest existing ancestor. Anc$_{\leq D}(g)$ is the list of the $D = 32$ nearest ancestors.
Demo of c	An episode prefix $\tau[0{:}t]$ from s_0 to the episode's first step in c with $R \geq R_c$. Each landmark keeps its top 4 demos ranked by (R desc., length asc.).
Return statistics of g	W_g: the last ≤64 return outcomes (1 = reached, 0 = failed or died), with succ(g) = mean(W_g). Also: ℓ_g, the EMA of successful return lengths; the counters n_g^{tgt} (episodes that targeted g), n_g^{reach}, n_g^{hand} (handoffs at g) and n_g^{qd} (handoffs followed by death within 8 steps).
ELIGIBLE(g)	The root is always eligible. Any other g is *ineligible* if ($n_g^{\text{tgt}} \geq 32$, $n_g^{\text{reach}} = 0$ and g has no demo), or if ($n_g^{\text{hand}} \geq 16$ and $n_g^{\text{qd}} > 0.9\, n_g^{\text{hand}}$). FALLBACK($g$) returns the first eligible landmark in $(g, \text{Anc}_1(g), \text{Anc}_2(g), \ldots)$, else c_0.
Slots, groups	$E = 128$ episode slots run in lockstep, in $M = 4$ groups $G_1..G_M$ of $N = 32$ slots. Group m has a goal g_m, the set I_m of landmarks its slots improved since its last consolidation, an episode counter n_m and a target K_m. Slot i has a next goal ν_i.
I	The landmarks improved (created or replaced) by the current episode.
π_R, V_R	Goal-conditioned actor-critic. Input: image x_t (4-frame stack, frame at the last reward, goal frame f_g; 6 channels) and context κ_t = ($\mathbf{1}$[Anc$_{\leq 32}(g)$ visited this episode], clip($(R_g - R_t)/1000, 0, 5$), $t/5000$). Separate IMPALA encoders for π_R and V_R.
β_t	Logit temperature (entropy boost): $\beta_t = 1 + (0.01 \max(0,\, t - t_{\text{prog}} - 100))^2$, where t_{prog} is the last step at which the episode entered a cell for the first time.

Algorithm 1′ RunEpisode (one episode of slot i; replaces one particle's outer step)

input: goal landmark $g \in \mathcal{A}$, return policy π_R, uncertainty U
params: return timeout $(\mu, b) = (2, 200)$, return stall 500, explore budget 1000, explore stall 300, episode cap $T_{\max} = 5000$, hold length $[h_{\min}, h_{\max}] = [5, 15]$, $\epsilon = 0.2$, quick death 8

$s \leftarrow \mathcal{E}.\textsc{reset}()$; $R \leftarrow 0$; $\tau \leftarrow (s)$; $I \leftarrow \emptyset$; $t \leftarrow 0$; $n_g^{\text{tgt}} \leftarrow n_g^{\text{tgt}} + 1$ // no restore: always from s_0
if $\text{cell}(s) = g$ and $R_g \le 0$ **then** // goal is the start cell (e.g. root)
  $n_g^{\text{reach}}, n_g^{\text{hand}} \leftarrow n_g^{\text{reach}}{+}1, n_g^{\text{hand}}{+}1$; $t_h \leftarrow 0$; **goto** line 21 // no return attempt is recorded
/* Return phase: replaces Reset(p_i, v) by walking to g */
$L \leftarrow \ell_g$ if defined, else length of g's best demo, else 1000
for $t = 1, 2, \ldots$ **do**
  $a_t \sim \pi_R(\cdot \mid x_t, \kappa_t)$ at temperature β_t
  $s, r, \mathit{dead} \leftarrow \mathcal{E}.\textsc{step}(a_t)$; $R \leftarrow R + r$; append (a_t, s) to τ
  if not *dead* **then** $I \leftarrow I \cup \textsc{UpdateArchive}(s, R, \tau, i)$ // Algorithm 2′
  $r_t^R \leftarrow \mathbf{1}[\textsc{Reach}(s, R, g) \wedge \neg \mathit{dead}] + 0.1 \cdot \#\{\text{ancestors in } \text{Anc}_{\le 32}(g) \text{ first visited at } t\} + 0.1\,\text{sign}(r)$
  if *dead* **then**
    $\textsc{RecordReturn}(g, 0)$; **return** $(\tau, I, \textsc{Dead})$ // Algorithm 3′
  else if $\textsc{Reach}(s, R, g)$ **then**
    $\textsc{RecordReturn}(g, 1)$; $\ell_g \leftarrow 0.9\,\ell_g + 0.1\,t$; $n_g^{\text{hand}} \leftarrow n_g^{\text{hand}} + 1$; $t_h \leftarrow t$; **break** // handoff
  else if $t \ge \mu L + b$ **or** $t - t_{\text{prog}} \ge 500$ **or** $t \ge T_{\max}$ **then**
    $\textsc{RecordReturn}(g, 0)$; $t_h \leftarrow \perp$ // failed: explore from here anyway
    if $t \ge T_{\max}$ **then return** $(\tau, I, \textsc{Alive})$
    break
/* Explore phase: GoWU's particle policy (held random action) */
$t_0 \leftarrow t$; $t_{\text{imp}} \leftarrow t$; $h \leftarrow 0$ // ≈ Alg. 1, l. 4--12
repeat
  if $h = 0$ **then** $a_{\text{hold}} \sim \text{Unif}(\mathcal{A}_{\text{act}})$; $h \sim \text{Unif}\{h_{\min}, \ldots, h_{\max}\}$
  $a_t \leftarrow a_{\text{hold}}$; $h \leftarrow h - 1$; with probability ϵ: $a_t \sim \text{Unif}(\mathcal{A}_{\text{act}})$
  $t \leftarrow t + 1$; $s, r, \mathit{dead} \leftarrow \mathcal{E}.\textsc{step}(a_t)$; $R \leftarrow R + r$; append (a_t, s) to τ
  if *dead* **then**
    if $t_h \ne \perp$ **and** $t - t_h \le 8$ **then** $n_g^{\text{qd}} \leftarrow n_g^{\text{qd}} + 1$ // feeds the doomed-landmark gate
    return $(\tau, I, \textsc{Dead})$
  $J \leftarrow \textsc{UpdateArchive}(s, R, \tau, i)$; $I \leftarrow I \cup J$; **if** $J \ne \emptyset$ **then** $t_{\text{imp}} \leftarrow t$
until $t - t_0 \ge 1000$ **or** $t - t_{\text{imp}} \ge 300$ **or** $t \ge T_{\max}$
return $(\tau, I, \textsc{Alive})$

Algorithm 2′ UpdateArchive (every Alive step; replaces $\mathcal{T}$.AddChild)

input: state s (frame f), episode reward R, trajectory $\tau = \tau[0{:}t]$, slot i in group m
params: U hysteresis $\eta = 0.1$, demos per landmark 4, demo store cap 12 GB

$c \leftarrow \text{cell}(s)$; $c_{\text{prev}} \leftarrow \text{cell}(s_{t-1})$
if $c \in \mathcal{A}$ **and** $R = R_c$ **then** $U_c \leftarrow U(f_c)$ // re-score the incumbent with the current RND
if $c \notin \mathcal{A}$ **or** $R > R_c$ **or** $(R = R_c$ **and** $U(f) > (1 + \eta)\,U_c)$ **then** // $(R, U) \succ^{\text{lex}} (R_c, U_c)$ with hysteresis
  if $c \notin \mathcal{A}$ **then**
    $u_c \leftarrow c_{\text{prev}}$ if $c_{\text{prev}} \ne c$, else none; $d_c \leftarrow d_{u_c} + 1$ (or 0) // parent fixed at creation
  else if $R > R_c$ **then**
    drop c's demos, W_c and ℓ_c // old routes no longer earn R_c
  $(f_c, R_c, U_c) \leftarrow (f, R, U(f))$; $J \leftarrow \{c\}$; $I_m \leftarrow I_m \cup \{c\}$ // I_m grows live, not at episode end
else
  $J \leftarrow \emptyset$
/* Hindsight demonstration */
if this is the first step of τ in c with $R \ge R_c$ **and** $t > 0$ **then**
  offer $\tau[0{:}t]$ as a demo of c; keep the top 4 by (R desc., t asc.) // only first demos once store is full
return J

Algorithm 3′ RecordReturn (after every return attempt; failure recovery via rollback)

input: landmark g, outcome $o \in \{0, 1\}$
params: window $w = 64$, success threshold $\rho = 0.05$, rollback range $[k_{\min}, k_{\max}] = [1, 3]$

append o to W_g (keep the last w); **if** $o = 1$ **then** $n_g^{\text{reach}} \leftarrow n_g^{\text{reach}} + 1$
if $|W_g| = w$ **and** $\text{succ}(g) < \rho$ **then** // ≈ Alg. 1, l. 14--17
  for each group m with $g_m = g$ **do**
    $k \sim \text{Unif}\{k_{\min}, \ldots, k_{\max}\}$; $g_m \leftarrow \text{Anc}_k(g)$; $\nu_j \leftarrow g_m$ for all $j \in G_m$ // see Remark 1
  $W_g \leftarrow \emptyset$ // cleared even if no group targets g

Algorithm 4′ LR-GoWU (replaces GoWU's population loop)

input: environment $\mathcal{E}$, $E = 128$ slots in $M = 4$ groups
params: consolidation $[K_{\min}, K_{\max}] = [16, 32]$ episodes, mixing $p_{\text{mix}} = 0.1$, unroll $T = 128$

Initialize $\mathcal{A} \leftarrow \{c_0\}$ with $(f_{c_0}, 0, U(f_{c_0}))$; π_R, V_R; shared RND U and its frame replay
$g_m \leftarrow c_0$, $I_m \leftarrow \emptyset$, $n_m \leftarrow 0$, $K_m \sim \text{Unif}\{K_{\min}..K_{\max}\}$ for all m; $\nu_i \leftarrow g_{m(i)}$ for all i
loop // all E slots step in lockstep
  for T steps **do**
    for each slot i in group m **in parallel do**
      if slot i has no running episode **then** // episode start
        $g \leftarrow \nu_i$; **if not** Eligible(g) **then** $g \leftarrow$ Fallback(g_m) // falls back from the group goal
        start RunEpisode(g) in slot i // Algorithm 1′
      advance slot i's episode by one environment step
      if the episode is still running **then continue**
      /* Episode end with (τ, I, status): choose the next goal */
      if $\text{Unif}(0, 1) < p_{\text{mix}}$ **then**
        $\nu_i \sim \{c \in \mathcal{A} : \text{Eligible}(c)\}$ with probability $\propto 1/\sqrt{n_c^{\text{tgt}} + 1}$ // count-based mixing
      else if status = Alive **and** $I \cap \text{Eligible} \neq \emptyset$ **then**
        $\nu_i \leftarrow \text{lex-argmax}_{c \in I \cap \text{Eligible}}(R_c, U_c)$ // survivor continues; ≈ Alg. 1, l. 10--11
      else
        $\nu_i \leftarrow \text{Winner}(m) := \text{lex-argmax}_{c \in \{g_m\} \cup (I_m \cap \text{Eligible})}(R_c, U_c)$ // ≈ Alg. 1, l. 23
      $n_m \leftarrow n_m + 1$
      if $n_m \geq K_m$ **then** // group consolidation; ≈ Alg. 1, l. 29--30
        $g_m \leftarrow \text{Winner}(m)$; $\nu_j \leftarrow g_m$ for all $j \in G_m$ // see Remark 1
        $I_m \leftarrow \emptyset$; $n_m \leftarrow 0$; $K_m \sim \text{Unif}\{K_{\min}..K_{\max}\}$
        $g^\star \leftarrow \text{lex-argmax}_{k \leq M}(R_{g_k}, U_{g_k})$ // global sync; ≈ Alg. 2, l. 8--10, 16
        for each group k with $R_{g_k} < R_{g^\star}$ **do**
          $g_k \leftarrow g^\star$; $\nu_j \leftarrow g^\star$ for all $j \in G_k$
  /* Learning, once per unroll of $T \cdot E$ = 16,384 agent steps */
  PPO on the unroll's return-phase steps only, reward r^R (skipped if fewer than 256)
    terminal at Reach or Dead; return timeout/stall bootstraps from $\gamma V_R(s_{t+1})$
  BC: 8 minibatches of 512 $(x_j, \kappa_j) \to a_j$ from demos: landmark $c \propto 1 - \text{succ}(c)$,
    skipping c with $|W_c| \geq 8$ and $\text{succ}(c) \geq 0.8$; demo uniform among c's; $j \sim \text{Unif}\{0..t_{\text{arr}}-1\}$
  RND: 31 updates per 1000 agent steps on batches of 128 frames from the 512k replay

Remark 1 (consolidation, sync and rollback only partly redirect slots). A slot's next goal ν_i is chosen when its episode ends (Alg. 4′, l. 12–18). The writes "$\nu_j \leftarrow g_m$" in consolidation, global sync and rollback therefore last only until slot j's own episode ends, which overwrites them. They take effect only for the slot that triggered the consolidation. Otherwise these operations act through g_m: it is part of the winner set of every dead slot, and it is the fallback for ineligible goals. Unlike GoWU (Alg. 1, l. 30), Alive slots that improved a landmark keep following their own frontier after a consolidation. After a rollback, the winner of a dead slot may still be a high-R landmark from I_m rather than the ancestor.

Remark 2 (where the counters change). n_g^{tgt} counts episodes at their start, after the eligibility fallback. W_g counts every return attempt, including deaths, but not the trivial handoff of Alg. 1′, l. 3. An R-improvement of g resets W_g, ℓ_g and the demos, but not n_g^{tgt}, n_g^{reach}, n_g^{hand} or n_g^{qd}.

Hyperparameters (v2)

Environment	Montezuma's Revenge, sticky actions 0.25, 18 actions, episode ends at the first life loss, cap 5000 steps
Archive	cell $(\text{level}, \text{room}, \#\text{keys}, \lfloor x/16 \rfloor, \lfloor y/16 \rfloor)$; U hysteresis 0.1; top-4 demos; 12 GB demo store
RND	Nature torso + 1024-1024 head, input-normalized, lr 3×10^{-4}, batch 128, 31 updates / 1k agent steps, replay 512k
π_R, V_R	IMPALA ResNet (16-32-32, 1024-d), separate for π_R and V_R; heads $1024{+}34 \to 512 \to 18$ / 1; bf16
PPO	$\gamma = 0.99$, $\lambda = 0.95$, lr 10^{-4}, clip 0.1, 1 epoch, 8 minibatches, entropy 10^{-3}, value 0.5, grad-norm 0.5
Return reward	1 on reaching g, 0.1 per new ancestor, $0.1\,\text{sign}(r)$
BC	alternating with PPO, 8×512 per update, cross-entropy, filter succ ≥ 0.8 after 8 attempts
Selection	$M = 4$, $N = 32$, $K \in [16, 32]$ episodes, $p_{\text{mix}} = 0.1$, rollback window 64, $\rho = 0.05$, $k \in [1, 3]$
Eligibility	unreachable gate 32 targets; doomed gate ≥ 16 handoffs with $> 90\%$ dying within 8 steps
Explore	held random action, hold $\in [5, 15]$, $\epsilon = 0.2$, budget 1000, stall 300

References

[1] Z. Mhammedi and J. Cohan. *Decoupling Exploration and Policy Optimization: Uncertainty Guided Tree Search for Hard Exploration.* arXiv:2603.22273, 2026.